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2027 CYCLE

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CUET UG

Chemistry

Electrochemistry

Official Syllabus Unit II · Class 12 · CUET UG 2027 Cycle Notes

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Syllabus Alignment

Sub-topic	Official Syllabus Pointer (NTA CUET UG Baseline)	Skills & Competencies Tested
Galvanic Cells & EMF	Electrochemical Cells, Galvanic Cells, Standard Electrode Potential, SHE as zero reference, EMF measurement ($E^0_{\text{cell}} = E^0_{\text{cat}} - E^0_{\text{an}}$).	Writing cell notation, identifying redox half-reactions, polarity of terminals, and computing standard cell EMF.
Nernst Equation & Thermodynamics	Nernst Equation (effect of ion concentration, temperature), calculation of cell potential, equilibrium constant (K_c), and Gibbs free energy ($\Delta_r G^0$).	Applying $E = E^0 - \left(\frac{0.0591}{n}\right)\log Q$, linking $\Delta G^0 = -nFE^0_{\text{cell}}$, and calculating K_c at 298 K.
Conductance & Kohlrausch's Law	Conductance (G), Conductivity (κ), Molar Conductivity (Λ_m), variation on dilution, Debye-Hückel-Onsager, Kohlrausch's Law of Independent Migration of Ions.	Distinguishing κ from Λ_m, explaining dilution curves, computing Λ^0_m of weak electrolytes, and calculating α and K_a.
Electrolysis & Faraday's Laws	Electrolytic cells, Faraday's 1st & 2nd Laws of Electrolysis, preferential discharge of ions, overpotential of oxygen.	Predicting anode/cathode products for molten vs aqueous salts, calculating mass/charge with $w = \frac{MIt}{nF}$.
Batteries, Fuel Cells & Corrosion	Primary batteries (dry cell, mercury cell), secondary batteries (lead-acid, Ni-Cd), $H_2 - O_2$ fuel cells, mechanism of corrosion and prevention (galvanisation).	Recognising anode/cathode reactions on discharge/recharge, fuel cell efficiency, and electrochemical rusting of iron.

EXAM PATTERN

CUET UG Exam Strategy: Unit II contributes 4 to 6 questions in the compulsory 50-question chemistry paper (+5, −1 mark scheme). The most frequently tested traps involve: (1) sign of electrodes (anode is negative in galvanic cells, but positive in electrolytic cells), (2) subtracting reduction potentials rather than adding them, (3) opposite dilution trends of κ (falls) and Λ_m (rises), and (4) overpotential dictating Cl_2 discharge over O_2 at the anode during brine electrolysis.

Concept Module 1: Electrochemical Cells & The Daniell Cell

An **electrochemical cell** is a device capable of either generating electrical energy from chemical reactions or using electrical energy to cause chemical reactions.

- **Galvanic (Voltaic) Cell:** Converts the Gibbs free energy of a **spontaneous** redox reaction ($\Delta G < 0$) into electrical work (e.g. Daniell cell, batteries).
- **Electrolytic Cell:** Uses external electrical energy to drive a **non-spontaneous** chemical reaction ($\Delta G > 0$) (e.g. electroplating, water electrolysis).

The Daniell Cell Architecture

The model galvanic cell couples a zinc half-cell with a copper half-cell:

- **Anode Half-Cell (Left):** Zinc rod dipped in 1.0 M $ZnSO_4$ solution.

$$Zn(s) \rightarrow Zn^{2+}(\text{aq}) + 2e^- \quad (E^0 = -0.76 \text{ V}, \text{Oxidation})$$

  The zinc electrode loses electrons and dissolves. Electrons accumulate on the rod, making the anode the **negative terminal** (LOAN: Left, Oxidation, Anode, Negative).

- **Cathode Half-Cell (Right):** Copper rod dipped in 1.0 M $CuSO_4$ solution.

$$Cu^{2+}(\text{aq}) + 2e^- \rightarrow Cu(s) \quad (E^0 = +0.34 \text{ V}, \text{Reduction})$$

  Copper ions consume incoming electrons and plate out as metallic copper. The cathode is the **positive terminal** (RED CAT: Reduction at Cathode).

- **Net Overall Cell Reaction:**

$$Zn(s) + Cu^{2+}(\text{aq}) \rightarrow Zn^{2+}(\text{aq}) + Cu(s) \quad (E^0_{\text{cell}} = +1.10 \text{ V})$$

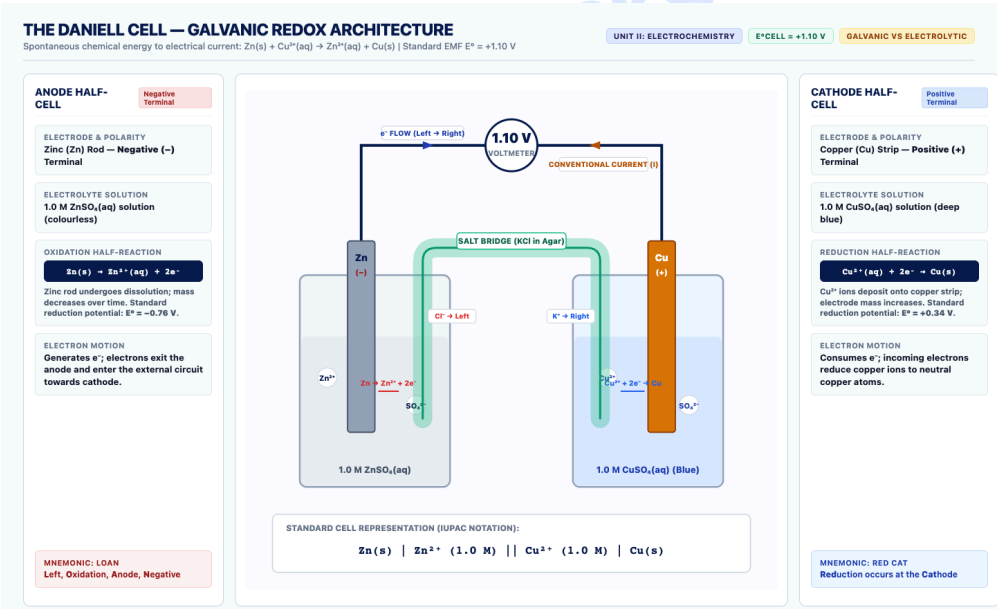


Figure 1: The Daniell Cell: Zn oxidation anode, Cu reduction cathode, porous salt bridge, and external electron flow through the voltmeter.

The Salt Bridge: Functions & Composition

A glass U-tube containing an inert electrolyte (KCl, KNO_3, or NH_4NO_3) set in agar-agar jelly.

- **Conditions for Salt Bridge Electrolyte:** The mobility and transport numbers of the cation (K^+) and anion (Cl^-) must be **virtually identical** to prevent liquid-junction potential.
- **Key Functions:**
  1. **Completes the electrical circuit** by allowing internal migration of ions.

2. **Maintains electrical neutrality** in both half-cell solutions (anions migrate to anode compartment to neutralise excess Zn^{2+}; cations migrate to cathode compartment to replace consumed Cu^{2+}).
3. **Eliminates Liquid Junction Potential** (the potential difference arising across the boundary of two dissimilar solutions).

IUPAC Cell Representation (Notation Rules)

By convention, the cell is written horizontally:

Anode Metal | Anode Electrolyte (conc.) ‖ Cathode Electrolyte (conc.) | Cathode Metal

For the Daniell cell under standard conditions:

$$Zn(s) \mid Zn^{2+}(1.0\text{ M}) \parallel Cu^{2+}(1.0\text{ M}) \mid Cu(s)$$

- Single vertical bar (|) denotes a phase boundary (solid electrode to liquid electrolyte).
- Double vertical line (‖) denotes the salt bridge.

Effect of External Opposing Potential (E_{ext}) on Daniell Cell

1. **When $E_{ext} < 1.10$ V:** The cell operates normally as a galvanic cell. Electrons flow from Zn to Cu; current flows from Cu to Zn. Zn dissolves and Cu deposits.
2. **When $E_{ext} = 1.10$ V:** Chemical equilibrium is reached. Net current drops to **zero**. No reaction occurs.
3. **When $E_{ext} > 1.10$ V:** The process is **reversed**! The cell operates as an **electrolytic cell**. Electrons flow from Cu to Zn; current flows from Zn to Cu. Zinc deposits on the zinc rod and copper dissolves from the copper rod.

The Salt Bridge: Construction, Mechanics & Functions

A salt bridge contains an agar-agar gel impregnated with an inert electrolyte (KCl, KNO_3, NH_4NO_3):

- **Equal Ionic Mobility:** Cations and anions must possess virtually identical transport numbers ($u_{K^+} \approx u_{Cl^-}$).
- **Elimination of Liquid-Junction Potential:** Equalises diffusion rates across the liquid boundary, eliminating interfacial voltage.
- **Electrical Neutrality:** Anions (Cl^-) diffuse to the anode to neutralize accumulating Zn^{2+}, while cations (K^+) migrate to the cathode to balance excess SO_4^{2-}.

Concept Module 2: Standard Electrode Potential & EMF

The potential difference between the electrode and its surrounding electrolyte solution is the **electrode potential**. When all species are at unit activity (1.0 M concentration, 1 bar gas pressure, 298 K), it is termed the **Standard Electrode Potential (E^0)**.

- By IUPAC convention, "electrode potential" without qualification strictly means **Standard Reduction Potential**.

Standard Hydrogen Electrode (SHE)

An isolated half-cell potential cannot be measured directly because every measurement requires a closed circuit. The **Standard Hydrogen Electrode (SHE)** serves as the universal zero reference:

$$Pt(s) \mid H_2(g, 1\text{ bar}) \mid H^+(aq, 1.0\text{ M}), \quad E^0 = 0.00\text{ V}$$

- Platinum foil coated with finely divided **platinum black** provides a catalytic surface for rapid equilibrium between H_2 gas and H^+ ions:

$$2H^+(aq) + 2e^- \leftrightarrow H_2(g)$$

Electromotive Force (EMF) Calculation

The cell EMF (E^0_{cell}) is the potential difference between cathode and anode when **no current flows** through the external circuit:

THE STANDARD EMF FORMULA

$$E^0_{\text{cell}} = E^0_{\text{cathode}} - E^0_{\text{anode}} = E^0_{\text{right}} - E^0_{\text{left}}$$

where both E^0_{cathode} and E^0_{anode} are standard **reduction** potentials from the electrochemical series.

- **Spontaneity Criterion:** If $E^0_{\text{cell}} > 0$, the redox reaction proceeds spontaneously in the forward direction.

The Electrochemical Series

Arranging redox couples in order of increasing reduction potentials yields the **Electrochemical Series**:

- **Strongest Reducing Agents (Lowest/Most Negative E^0):** Lithium ($E^0 = -3.05$ V), Potassium (-2.93 V), Sodium (-2.71 V), Zinc (-0.76 V). Readily lose electrons (undergo oxidation).
- **Zero Reference:** Hydrogen ($E^0 = 0.00$ V).
- **Strongest Oxidising Agents (Highest/Most Positive E^0):** Copper ($+0.34$ V), Silver ($+0.80$ V), Chlorine ($+1.36$ V), Fluorine ($+2.87$ V). Readily gain electrons (undergo reduction).
- **Displacement Rule:** A metal with lower E^0 (more negative) can displace a metal with higher E^0 from its aqueous salt solution:

$$Zn(E^0 = -0.76\text{ V}) + Cu^{2+}(E^0 = +0.34\text{ V}) \rightarrow Zn^{2+} + Cu \quad \left(E^0_{\text{cell}} = +1.10\text{ V} > 0\right)$$

Strategic Applications of the Electrochemical Series

1. **Oxidising vs Reducing Strength:** Elements with the most negative standard potentials ($\frac{\text{Li}^+}{\text{Li}} = -3.05$ V) are the **strongest reducing agents**. Elements with the most positive potentials ($\frac{\text{F}_2}{\text{F}^-} = +2.87$ V) are the **strongest oxidising agents**.
2. **Spontaneity Criterion:** A redox reaction is spontaneous under standard conditions if $E^0_{\text{cell}} = E^0_{\text{cathode}} - E^0_{\text{anode}} > 0$ ($\Delta G^0 < 0$).
3. **Hydrogen Displacement:** Metals with negative reduction potentials (Zn, Fe, Mg) displace H_2 from dilute non-oxidising acids; metals with positive potentials (Cu, Ag) cannot.

Concept Module 3: Nernst Equation & Thermodynamics

Walther Nernst established the mathematical relationship between the electrode potential and the concentration of electrolytic species at any temperature.

Nernst Equation for a General Half-Cell Reaction

For a general reduction half-reaction:

$$M^{n+}(\text{aq}) + ne^- \rightarrow M(s)$$

$$E_{\left(\frac{M^{n+}}{M}\right)} = E^0_{\left(\frac{M^{n+}}{M}\right)} - \frac{RT}{nF} \ln \frac{1}{[M^{n+}]}$$

Substituting universal constants at $T = 298$ K ($R = 8.314$ J K^{-1}mol^{-1}, $F = 96487$ C mol^{-1}, conversion factor 2.303):

$$E_{\left(\frac{M^{n+}}{M}\right)} = E^0_{\left(\frac{M^{n+}}{M}\right)} - \frac{0.0591}{n} \log \frac{1}{[M^{n+}]} = E^0_{\left(\frac{M^{n+}}{M}\right)} + \frac{0.0591}{n} \log[M^{n+}]$$

Nernst Equation for a Complete Galvanic Cell

For an arbitrary redox reaction: $aA + bB \rightarrow cC + dD$

$$E_{\text{cell}} = E^0_{\text{cell}} - \frac{0.0591}{n} \log Q = E^0_{\text{cell}} - \frac{0.0591}{n} \log \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

where n is the number of moles of electrons transferred in the balanced cell equation, and Q is the **Reaction Quotient**.

- For the Daniell Cell ($n = 2$):

$$E_{\text{cell}} = 1.10 - \frac{0.0591}{2} \log \frac{[Zn^{2+}]}{[Cu^{2+}]}$$

  - Increasing $[Cu^{2+}]$ (reactant) **increases** E_{cell}.
  - Increasing $[Zn^{2+}]$ (product) **decreases** E_{cell}.

Equilibrium Constant (K_c) from Nernst Equation

As a galvanic cell operates, reactants are consumed while products accumulate. Consequently, E_{cell} gradually falls until reaching **zero** ($E_{\text{cell}} = 0$). At this stage, the reaction reaches dynamic chemical equilibrium ($Q = K_c$):

$$0 = E^0_{\text{cell}} - \frac{2.303RT}{nF} \log K_c$$

$$\log K_c = \frac{n \times E^0_{\text{cell}}}{0.0591} \quad (\text{at } 298 \text{ K})$$

For the Daniell cell ($n = 2, E^0_{\text{cell}} = 1.10$ V):

$$\log K_c = \frac{2 \times 1.10}{0.0591} = 37.22 \Longrightarrow K_c \approx 2 \times 10^{37}$$

Gibbs Free Energy of Reaction ($\Delta_r G$)

The maximum electrical work (W_{max}) done by a galvanic cell equals the decrease in its Gibbs free energy:

$$W_{\text{max}} = q \times E_{\text{cell}} = nFE_{\text{cell}}$$

$$\Delta_r G = -nFE_{\text{cell}} \quad \text{and} \quad \Delta_r G^0 = -nFE^0_{\text{cell}}$$

- **Thermodynamic Relationships:**
  - If $E^0_{\text{cell}} > 0 \Longrightarrow \Delta_r G^0 < 0$ (Reaction is spontaneous).
  - If $E^0_{\text{cell}} < 0 \Longrightarrow \Delta_r G^0 > 0$ (Reaction is non-spontaneous).
  - Also: $\Delta_r G^0 = -RT \ln K_c = -2.303RT \log K_c$.

Concentration Cells & Temperature Coefficient of EMF

1. **Electrolyte Concentration Cells:** Identical electrodes dipped in solutions of different concentrations ($C_1 < C_2$):

$$E^0_{\text{cell}} = 0 \Longrightarrow E_{\text{cell}} = \left(\frac{0.0591}{n}\right) \log\left(\frac{C_2}{C_1}\right) \quad (C_2 > C_1)$$

2. **Temperature Coefficient of EMF:** $\left(\partial \frac{E}{\partial} T\right)_p = \frac{\Delta S}{nF}$ and $\Delta H = nF\left[T\left(\partial \frac{E}{\partial} T\right)_p - E\right]$.

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Concept Module 4: Electrolytic Conductance & Dilution Dynamics

Electric current passes through solutions via the movement of hydrated ions (**electrolytic conduction**). Unlike metallic conduction, electrolytic conduction involves actual transfer of matter and its conductance **increases** with rising temperature (due to reduced viscosity and increased kinetic energy of ions).

Fundamental Conductance Parameters

1. **Resistance (R):** Governed by Ohm's Law: $R = \rho \times \frac{l}{A}$, measured in Ohms ($\Omega$).
2. **Conductance (G):** Reciprocal of resistance: $G = \frac{1}{R}$, measured in Siemens (S) or Ω^{-1} (mho).
3. **Resistivity (ρ):** Resistance of a conductor having unit length and unit cross-sectional area: $\rho = R \times \frac{A}{l}$ (Ω cm).
4. **Conductivity (κ, Specific Conductance):** Reciprocal of resistivity:

$$\kappa = \frac{1}{\rho} = \frac{1}{R} \times \frac{l}{A} = G \times G^*$$

   where $G^* = \frac{l}{A}$ is the **Cell Constant** (expressed in cm^{-1} or m^{-1}). Units of κ: S cm^{-1} or S m^{-1}.
5. **Molar Conductivity (Λ_m):** The conducting power of all ions produced by dissolving 1 mole of an electrolyte in solution:

$$\Lambda_m = \frac{1000 \times \kappa}{C} \quad [\text{in S cm}^2\ \text{mol}^{-1}]$$

   where C is molar concentration (mol/L). (In SI units: $\Lambda_m = \frac{\kappa}{1000 \times C}$ in S m^2 mol^{-1}).

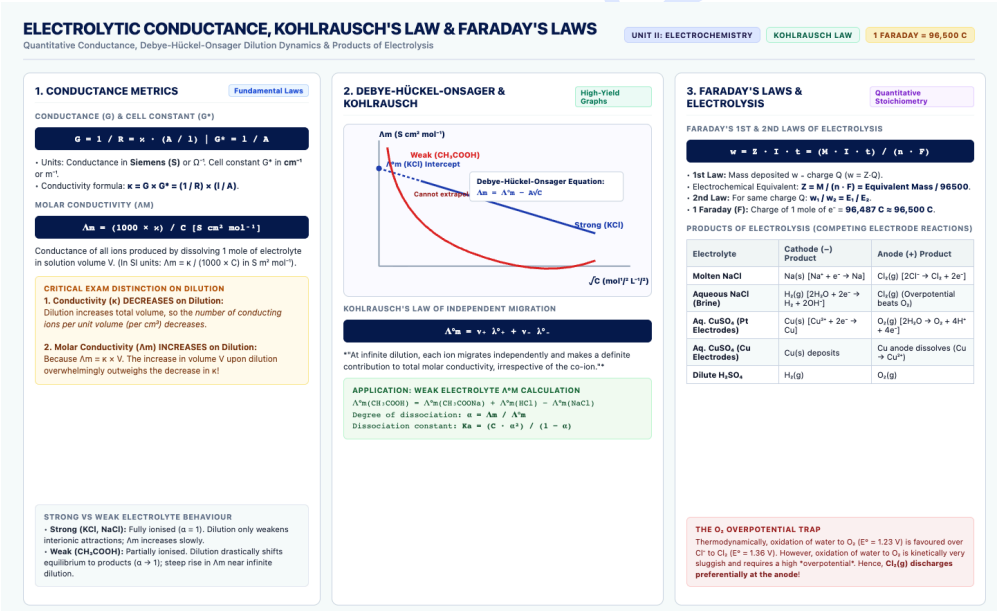


Figure 2: Electrolytic Conductance & Kohlrausch's Law: Dilution dynamics of strong vs weak electrolytes, Debye-Hückel-Onsager plot, and Faraday's electrolysis stoichiometry.

Effect of Dilution on Conductivity (κ) vs Molar Conductivity (Λ_m)

This is the single most tested theoretical distinction in CUET UG Chemistry:

1. **Conductivity (κ) DECREASES with Dilution:** κ is the conductance of 1 cm^3 of solution. As a solution is diluted, the total volume expands, meaning the **number of conducting ions per unit volume (1 cm^3) decreases**. Hence, κ always drops upon dilution for both strong and weak electrolytes.

2. **Molar Conductivity (Λ_m) INCREASES with Dilution:** $\Lambda_m = \kappa \times V$, where V is the volume containing 1 mole of electrolyte. While κ decreases upon dilution, V increases by a far greater factor. The product $\kappa \times V$ increases, so Λ_m always rises upon dilution.

Debye-Hückel-Onsager Equation for Strong Electrolytes

Strong electrolytes ($KCl, NaCl, HCl$) dissociate completely even at moderate concentrations ($\alpha = 1$). Dilution only separates ions further apart, diminishing interionic attractions. The variation follows the linear **Debye-Hückel-Onsager equation**:

$$\Lambda_m = \Lambda_m^0 - A\sqrt{C}$$

- Plotting Λ_m against $\sqrt{C}$ yields a straight line with:
  - y-intercept $= \Lambda_m^0$ (**Limiting Molar Conductivity** at infinite dilution, $C \to 0$).
  - Slope $= -A$ (constant depending on solvent, temperature, and electrolyte charges).
- Extrapolation to $\sqrt{C} = 0$ reliably gives Λ_m^0 for strong electrolytes.

Weak Electrolytes (CH_3COOH, NH_4OH)

Weak electrolytes have low degree of dissociation ($\alpha \ll 1$). On dilution, α increases according to Ostwald's Dilution Law ($\alpha \approx \sqrt{\frac{K_a}{C}}$).

- Near infinite dilution ($C \to 0$), α rapidly approaches 1, causing an **exponential asymptotic steep rise** in Λ_m.
- The curve runs nearly parallel to the y-axis and **cannot be extrapolated** to intercept the axis. Hence, Λ_m^0 of weak electrolytes **cannot** be obtained experimentally from a Debye-Hückel plot!

Concept Module 5: Kohlrausch's Law & Faraday's Laws

Kohlrausch's Law of Independent Migration of Ions

To determine Λ_m^0 for weak electrolytes, Friedrich Kohlrausch formulated:

KOHLRAUSCH'S LAW FORMULATION

"At infinite dilution, when dissociation is complete and interionic attractions vanish, each ion makes a definite and independent contribution towards the molar conductivity of the electrolyte, regardless of the nature of the co-ion."

$$\Lambda_m^0 = \nu_+ \lambda_+^0 + \nu_- \lambda_-^0$$

where λ_+^0 and λ_-^0 are the limiting ionic conductivities of the cation and anion, and ν_+, ν_- are the stoichiometric coefficients.

Practical Calculations with Kohlrausch's Law

1. **Determining Λ_m^0 of Acetic Acid:**

$$\Lambda_m^0(CH_3COOH) = \Lambda_m^0(CH_3COONa) + \Lambda_m^0(HCl) - \Lambda_m^0(NaCl)$$

2. **Degree of Dissociation (α):**

$$\alpha = \frac{\Lambda_m}{\Lambda_m^0}$$

3. **Dissociation Constant (K_a):**

$$K_a = \frac{C\alpha^2}{1-\alpha} = \frac{C\left(\frac{\Lambda_m}{\Lambda_m^0}\right)^2}{1 - \frac{\Lambda_m}{\Lambda_m^0}}$$

Faraday's Laws of Electrolysis

Michael Faraday established the quantitative relationship between electrical charge and chemical change:

1. **Faraday's First Law:** The mass (w) of substance deposited or liberated at any electrode is directly proportional to the quantity of electricity (Q) passed:
$$w \propto Q \Longrightarrow w = Z \times Q = Z \times I \times t$$
where Z is the **Electrochemical Equivalent**:
$$Z = \frac{\text{Molar Mass } M}{n \times F} = \frac{\text{Equivalent Mass } E}{96500}$$
$$w = \frac{M \times I \times t}{n \times F}$$
2. **Faraday's Second Law:** When the same quantity of electricity is passed through different electrolyte solutions connected in series, the masses of substances liberated are directly proportional to their chemical equivalent masses (E):
$$\frac{w_1}{w_2} = \frac{E_1}{E_2} = \frac{\frac{M_1}{n_1}}{\frac{M_2}{n_2}}$$
3. **The Faraday Constant (F):** The charge carried by 1 mole of electrons:
$$1 \text{ F} = N_A \times e = (6.022 \times 10^{23}) \times (1.602 \times 10^{-19} \text{ C}) = 96487 \text{ C} \approx 96500 \text{ C mol}^{-1}$$

Competing Products of Electrolysis

When multiple ions are present in solution, preferential discharge depends on electrode potential, concentration, and **overpotential**:

- **Cathode Reduction Competition:** The redox couple with the **higher (more positive) reduction potential** discharges first.
  - In aqueous $NaCl$ (brine): $E^0\left(N\frac{a^+}{N}a\right) = -2.71$ V; $E^0\left(H_2\frac{O}{H_2}\right) = -0.83$ V. Water reduces preferentially:
$$2H_2O + 2e^- \rightarrow H_2(g) + 2OH^-(\text{aq}) \quad \Longrightarrow H_2(g) \text{ is evolved at cathode.}$$
- **Anode Oxidation Competition & The Overpotential Anomaly:**
  - In aqueous $NaCl$: Oxidation of water to O_2 has $E^0 = +1.23$ V, while oxidation of Cl^- to Cl_2 has $E^0 = +1.36$ V.
  - Thermodynamically, water oxidation is favoured. However, the release of O_2 involves breaking multiple bonds and is kinetically slow, requiring a significant extra potential (**overpotential**). Consequently, **chlorine gas (Cl_2) discharges preferentially at the anode**!

Applications of Kohlrausch's Law & Products of Electrolysis

1. **Weak Electrolyte Ionization:** Degree of dissociation $\alpha = \frac{\Lambda_m}{\Lambda_m^0}$ and $K_a = \frac{c\alpha^2}{1-\alpha}$.
2. **Sparingly Soluble Salts:** Solubility $S = \frac{\kappa \times 1000}{\Lambda_m^0}$ (mol L^{-1}) and $K_{\text{sp}} = S^2$ for 1 : 1 binary salts (AgCl).
3. **Electrolysis Products (Overpotential):** In aqueous brine (NaCl), H_2 gas evolves at cathode ($E^0(H^+) > E^0(\text{Na}^+)$) and Cl_2 gas evolves at anode (overpotential of oxygen makes Cl_2 discharge kinetically faster).

Concept Module 6: Commercial Cells & Corrosion

Commercial batteries are compact, portable galvanic cells connected in series.

1. Primary Batteries (Non-Rechargeable)

The redox reaction occurs only once; after use, the battery dies and cannot be reused.

- **Dry Cell (Leclanché Cell):**
  - **Anode:** Zinc cylinder container: $Zn(s) \rightarrow Zn^{2+} + 2e^-$.
  - **Cathode:** Graphite rod surrounded by powdered MnO_2 and carbon.
  - **Electrolyte:** Moist paste of NH_4Cl and $ZnCl_2$.
  - **Cathode Reaction:** $MnO_2 + NH_4^+ + e^- \rightarrow MnO(OH) + NH_3$.
  - Complex formation: $Zn^{2+} + 4NH_3 \rightarrow \left[Zn(NH_3)_4\right]^{2+}$ prevents pressure buildup from free ammonia. Cell potential: 1.25 V to 1.5 V.
- **Mercury Cell (Hearing aids, watches):**
  - **Anode:** Zinc-mercury amalgam ($Zn - Hg$).
  - **Cathode:** Paste of mercuric oxide (HgO) and carbon.
  - **Electrolyte:** Moist paste of KOH and ZnO.
  - **Overall Reaction:** Zn(amalgam) $+ HgO(s) \rightarrow ZnO(s) + Hg(l)$.
  - **Key Exam Feature:** Delivers a constant voltage of 1.35 V throughout its lifetime because the overall reaction involves **no ions in solution** whose concentrations can change.

2. Secondary Batteries (Rechargeable)

Can be recharged by passing direct current in the opposite direction, acting as an electrolytic cell during charging.

- **Lead Storage Battery (Automobiles, Inverters):**
  - **Anode:** Spongy lead plates (Pb).
  - **Cathode:** Grid of lead packed with lead dioxide (PbO_2).
  - **Electrolyte:** 38%$\left(\frac{w}{w}\right)$ sulphuric acid (H_2SO_4, density 1.30 g/mL).
  - **Discharging Reactions:**
    - Anode: $Pb(s) + SO_4^{2-}(\text{aq}) \rightarrow PbSO_4(s) + 2e^-$
    - Cathode: $PbO_2(s) + 4H^+ + SO_4^{2-} + 2e^- \rightarrow PbSO_4(s) + 2H_2O(l)$
    - Overall: $Pb(s) + PbO_2(s) + 2H_2SO_4(\text{aq}) \rightarrow 2PbSO_4(s) + 2H_2O(l)$
    - Both electrodes become coated with white insoluble $PbSO_4$ and H_2SO_4 is consumed (density falls below 1.20 g/mL).
  - **Recharging Reaction:** Exactly reverses the above: $2PbSO_4(s) + 2H_2O \rightarrow Pb(s) + PbO_2(s) + 2H_2SO_4$.
- **Nickel-Cadmium Cell:** Cadmium anode and $NiO(OH)$ cathode. Longer cycle life than lead-acid, but heavier and more expensive. Cell voltage 1.4 V.

3. Fuel Cells

Galvanic cells designed to continuously convert the combustion energy of fuels (H_2, CH_4, CH_3OH) into electricity.

- **Hydrogen-Oxygen ($H_2 - O_2$) Fuel Cell (Apollo Spacecraft):**
  - **Electrodes:** Porous carbon impregnated with catalysts (Pt, Pd).
  - **Electrolyte:** Concentrated aqueous KOH or $NaOH$.
  - **Anode:** $2H_2(g) + 4OH^-(\text{aq}) \rightarrow 4H_2O(l) + 4e^-$
  - **Cathode:** $O_2(g) + 2H_2O(l) + 4e^- \rightarrow 4OH^-(\text{aq})$
  - **Net Reaction:** $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$.
  - **Efficiency:** Runs at **70% efficiency** (compared to 40% for thermal plants). Zero greenhouse emissions; water produced was condensed and used for astronaut drinking supply.

4. Corrosion of Metals (Rusting of Iron)

An electrochemical phenomenon where metal surfaces undergo oxidation:

- **Anodic Spot:** Pure iron oxidises: $2Fe(s) \rightarrow 2Fe^{2+} + 4e^{-} (E^0 = -0.44 \text{ V})$.
- **Cathodic Spot:** Electrons travel through the metal to another spot where dissolved O_2 in the presence of H^+ (from atmospheric $CO_2 + H_2O \leftrightarrow H_2CO_3$) reduces:

$$O_2(g) + 4H^+(\text{aq}) + 4e^- \rightarrow 2H_2O(l) \quad (E^0 = +1.23 \text{ V})$$

- **Rust Formation:** Fe^{2+} ions are further oxidised by atmospheric oxygen to hydrated ferric oxide ($Fe_2O_3 \cdot xH_2O$, rust).
- **Prevention:**
  1. **Barrier Protection:** Painting, oiling, or grease coating.
  2. **Sacrificial Protection (Galvanisation):** Coating iron with a thin layer of **Zinc** ($E^0 = -0.76$ V). Because Zinc has a more negative reduction potential than iron ($E^0 = -0.44$ V), Zinc corrodes preferentially even when scratched!

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Figure 3: Unit II Electrochemistry Master Architectural Mindmap: Integrated overview of galvanic cells, thermodynamic equations, electrolytic conductance dynamics, Kohlrausch's law, Faraday's electrolysis laws, and commercial batteries.

MASTER ELECTROCHEMICAL ARCHITECTURE & CELL DYNAMICS BREAKDOWN

The cognitive map above unifies the mathematical foundations and core principles across Unit II:

- **Galvanic vs Electrolytic Regimes:** Trace the behavior of the Daniell cell under external voltage: when $E_{\text{ext}} < 1.10$ V, the cell operates spontaneously (galvanic); when $E_{\text{ext}} = 1.10$ V, net current is zero; when $E_{\text{ext}} > 1.10$ V, electron flow reverses and the cell is driven as an electrolytic cell.
- **Nernst Equation Scaling:** Remember that cell potential is an intensive property (E^0 is independent of stoichiometric multiplication), while $\Delta G^0 = -nFE^0$ is extensive. Always confirm the number of transferred electrons n and write the reaction quotient Q with correct stoichiometric exponents.
- **Conductance on Dilution Paradox:** Always distinguish between conductivity (κ) and molar conductivity (Λ_m). Upon dilution, κ drops universally due to fewer charge carriers per unit volume, while Λ_m rises for both strong and weak electrolytes.
- **Commercial Battery Architectures:** Note why mercury cells deliver an unwavering 1.35 V potential—their overall cell reaction involves zero aqueous ions whose concentrations could vary during discharge.

UNIT II MASTER ELECTROCHEMISTRY FORMULA & HIGH-YIELD CHEATSHEET

Core Concept / Law	Mathematical Expression	Application & CUET Problem Strategy
Standard Cell EMF	$E^0_{\text{cell}} = E^0_{\text{cathode}} - E^0_{\text{anode}}$	Use standard reduction potentials strictly. Positive E^0 denotes spontaneity.
Nernst Equation (298 K)	$E_{\text{cell}} = E^0_{\text{cell}} - \left(\frac{0.0591}{n}\right)\log Q$	Pure solids and liquids have activity = 1. n is total moles of electrons transferred.
Equilibrium Constant	$\log K_c = \frac{nE^0_{\text{cell}}}{0.0591}$	At equilibrium, $E_{\text{cell}} = 0$ and $Q = K_c$. Relates electrochemistry to equilibrium.
Gibbs Free Energy	$\Delta G^0 = -nFE^0_{\text{cell}} = -2.303RT\log K_c$	For spontaneous process, $\Delta G^0 < 0$ when $E^0 > 0$. $F = 96487 \approx 96500\text{C mol}^{-1}$.
Conductivity (κ)	$\kappa = G \times G^* = \left(\frac{1}{R}\right) \times \left(\frac{l}{A}\right)$	G^* is cell constant (cm^{-1}), determined using standard 0.1 M KCl solutions.
Molar Conductivity	$\Lambda_m = \frac{\kappa \times 1000}{M}$	Units: S cm^2 mol^{-1}. Increases with dilution for all electrolytes.
Debye-Hückel-Onsager	$\Lambda_m = \Lambda^0_m - A\sqrt{c}$	Applies to strong electrolytes at low concentration. Interionic attractions weaken on dilution.
Kohlrausch's Law	$\Lambda^0_m = \nu_+\lambda^0_+ + \nu_-\lambda^0_-$	Enables calculation of Λ^0_m for weak electrolytes via additivity of ionic conductances.
Degree of Dissociation	$\alpha = \frac{\Lambda_m}{\Lambda^0_m} \mid K_a = \frac{c\alpha^2}{1-\alpha}$	Direct determination of weak electrolyte ionization and acid dissociation constant.
Faraday's First Law	$w = z \times I \times t = \left(\frac{M}{nF}\right) \times I \times t$	$z = \frac{M}{nF}$ is electrochemical equivalent. Charge $Q = I \times t$ Coulombs.
Faraday's Second Law	$\frac{w_1}{E_1} = \frac{w_2}{E_2} = \frac{Q}{F}$	Same quantity of electricity produces masses in direct ratio of their equivalent weights.
Mercury Cell Potential	$E_{\text{cell}} = 1.35$ V (Constant throughout)	Overall reaction $\text{Zn(Hg)} + \text{HgO}(s) \rightarrow \text{ZnO}(s) + \text{Hg}(l)$ involves no solution ions.

TOP 7 HIGH-FREQUENCY CUET TRAPS & PITFALLS

1. **External Voltage Regimes on Daniell Cell:** When $E_{\text{ext}} < 1.10$ V, electrons flow Zn → Cu (galvanic). When $E_{\text{ext}} = 1.10$ V, zero current flows (equilibrium). When $E_{\text{ext}} > 1.10$ V, polarity reverses! Electrons flow Cu → Zn, Zn deposits, and the device functions as an **electrolytic cell**.
2. **Reduction Potential Sign Convention:** All IUPAC standard potentials are reduction potentials. A more negative E^0 indicates a **stronger reducing agent** ($\frac{\text{Li}^+}{\text{Li}} = -3.05$ V is strongest reducing agent). A more positive E^0 indicates a stronger oxidising agent ($\frac{\text{F}_2}{\text{F}^-} = +2.87$ V).
3. **Nernst Equation Stoichiometry:** In $E = E^0 - \left(\frac{0.0591}{n}\right)\log Q$, concentrations of pure solids ($\text{Zn}(s), \text{Cu}(s)$) and pure liquids ($\text{H}_2$ O) are strictly taken as unity (1). Never insert solid concentrations into Q! Always balance the redox reaction to find n.
4. **Thermodynamic Properties (Intensive vs Extensive):** E^0_{cell} is an **intensive property**—multiplying a half-reaction by 2 does NOT change E^0. However, Gibbs free energy $\Delta G^0 = -nFE^0$ is an **extensive property**—doubling the stoichiometric equation doubles ΔG^0!
5. **Conductivity (κ) vs Molar Conductivity (Λ_m) on Dilution:** Upon dilution, conductivity (κ) **decreases** because the number of current-carrying ions per unit volume (1cm^3) decreases. However, molar conductivity (Λ_m) **increases** on dilution for both strong and weak electrolytes!
6. **Faraday's Electricity & Electrolysis Products:** Discharge of 1 mole of an n-valent ion requires nF ($n \times 96500$ C). In electrolysis of brine (aqueous NaCl), H_2 gas evolves at the cathode (since $E^0(H^+) > E^0(\text{Na}^+)$) and Cl_2 gas evolves at the anode (due to the overpotential of oxygen).
7. **Mercury Cell vs Lead Storage Accumulator:** Mercury cells maintain an unvarying potential of 1.35 V throughout their life because their overall cell reaction involves **no ions in solution** whose concentration can change. In lead storage batteries, sulphuric acid is consumed during discharge (density falls from 1.30 to < 1.20 g cm^{-3}).

Practice Questions (Exam Pattern) — Part 1 (Q1 to Q5)

Q1. In a operating Daniell cell ($Zn \mid Zn^{2+} \parallel Cu^{2+} \mid Cu$), which statement correctly describes the electrodes and electron flow?

(A) Zinc is the negative anode, copper is the positive cathode, and electrons flow from zinc to copper
(B) Zinc is the cathode, copper is the anode, and electrons flow from copper to zinc
(C) Copper is the negative anode, zinc is the positive cathode, and electrons flow from zinc to copper
(D) Zinc is the positive anode, copper is the negative cathode, and current flows from zinc to copper

Sol. Q1. Oxidation occurs at the zinc anode (negative terminal, LOAN), releasing electrons that travel through the external circuit to the copper cathode (positive terminal).

Answer: (A)

Q2. Given standard reduction potentials: $E^0\left(F\frac{e^{2+}}{F}e\right) = -0.44$ V and $E^0\left(C\frac{r^{3+}}{C}r\right) = -0.74$ V. The standard EMF of the cell $Cr \mid Cr^{3+} \parallel Fe^{2+} \mid Fe$ is:

(A) -1.18 V
(B) $+0.30$ V
(C) -0.30 V
(D) $+1.18$ V

Sol. Q2. $E^0_{\text{cell}} = E^0_{\text{cathode}} - E^0_{\text{anode}} = E^0\left(F\frac{e^{2+}}{F}e\right) - E^0\left(C\frac{r^{3+}}{C}r\right) = -0.44 - (-0.74) = +0.30$ V.

Answer: (B)

Q3. What is the standard hydrogen electrode potential (E^0) by international convention, and what is its composition?

(A) $E^0 = 0.00$ V, with H_2 at 1 bar bubbled over platinised Pt in 1.0 M H^+
(B) $E^0 = 1.00$ V, with H_2 at 1 atm bubbled over copper foil in 1 M HCl
(C) $E^0 = -0.76$ V, with H_2 gas dissolved in saturated KCl agar gel
(D) $E^0 = 0.059$ V, with pure H_2O at 298 K under 1 atm pressure

Sol. Q3. SHE is assigned $E^0 = 0.00$ V at all temperatures by IUPAC convention: $Pt \mid H_2(g, 1\text{ bar}) \mid H^+(\text{aq}, 1\text{ M})$.

Answer: (A)

Q4. For the cell reaction $Zn(s) + Cu^{2+}(\text{aq}) \rightarrow Zn^{2+}(\text{aq}) + Cu(s)$, the EMF of the cell decreases when:

(A) The concentration of Cu^{2+} ions is increased
(B) The concentration of Zn^{2+} ions is increased
(C) The surface area of the zinc electrode is doubled
(D) The volume of the salt bridge is increased

Sol. Q4. From Nernst equation: $E = E^0 - \left(\frac{0.059}{2}\right)\log\left(\frac{[Zn^{2+}]}{[Cu^{2+}]}\right)$. Increasing product $[Zn^{2+}]$ raises the subtracted log term, decreasing E_{cell}.

Answer: (B)

Q5. The standard cell potential (E^0_{cell}) for a 2-electron transfer redox reaction is $+0.2955$ V at 298 K. The equilibrium constant (K_c) for the reaction is:

(A) 1×10^{10}
(B) 1×10^{5}
(C) 2×10^{10}
(D) 1×10^{-10}

Sol. Q5. $\log K_c = \frac{nE^0_{\text{cell}}}{0.0591} = \frac{2 \times 0.2955}{0.0591} = 10.0 \Longrightarrow K_c = 10^{10} = 1 \times 10^{10}$.

Answer: (A)

Practice Questions (Exam Pattern) — Part 2 (Q6 to Q10)

Q6. How do the conductivity (κ) and molar conductivity (Λ_m) of an electrolytic solution vary upon dilution?

(A) Both κ and Λ_m increase upon dilution
(B) Both κ and Λ_m decrease upon dilution
(C) κ decreases while Λ_m increases upon dilution
(D) κ increases while Λ_m decreases upon dilution

Sol. Q6. Conductivity (κ) decreases because ion count per cm^3 falls. Molar conductivity ($\Lambda_m = \kappa V$) increases because volume V expands by a greater factor.

Answer: (C)

Q7. According to the Debye-Hückel-Onsager equation ($\Lambda_m = \Lambda_m^0 - A\sqrt{C}$), a plot of Λ_m vs $\sqrt{C}$ for potassium chloride (KCl) yields:

(A) A horizontal line parallel to the concentration axis
(B) A straight line with negative slope ($-A$) intercepting at Λ_m^0
(C) An exponential curve that runs parallel to the molar conductivity axis
(D) A parabola passing directly through the coordinate origin

Sol. Q7. For strong electrolytes (KCl), Λ_m decreases linearly with $\sqrt{C}$, giving intercept Λ_m^0 and negative slope $-A$.

Answer: (B)

Q8. Given: $\Lambda_m^0(NaCl) = 126.4$, $\Lambda_m^0(HCl) = 425.9$, and $\Lambda_m^0(CH_3COONa) = 91.0$ S cm^2 mol^{-1}. The Λ_m^0 for acetic acid is:

(A) 516.9 S cm^2 mol^{-1}
(B) 208.5 S cm^2 mol^{-1}
(C) 390.5 S cm^2 mol^{-1}
(D) 643.3 S cm^2 mol^{-1}

Sol. Q8. Kohlrausch's law: $\Lambda_m^0(CH_3COOH) = \Lambda_m^0(CH_3COONa) + \Lambda_m^0(HCl) - \Lambda_m^0(NaCl) = 91.0 + 425.9 - 126.4 = 390.5$ S cm^2 mol^{-1}.

Answer: (C)

Q9. The molar conductivity of 0.025 mol L^{-1} methanoic acid is 46.1 S cm^2 mol^{-1}. If $\Lambda_m^0 = 400.0$ S cm^2 mol^{-1}, its degree of dissociation (α) is:

(A) 0.115
(B) 0.230
(C) 0.058
(D) 0.869

Sol. Q9. $\alpha = \frac{\Lambda_m}{\Lambda_m^0} = \frac{46.1}{400.0} = 0.11525 \approx 0.115$ (or 11.5%).

Answer: (A)

Q10. How much charge in Coulombs is required to deposit all copper from 500 mL of a 0.20 M $CuSO_4$ solution via electrolysis?

(A) 9,650 C
(B) 19,300 C
(C) 38,600 C
(D) 48,250 C

Sol. Q10. Moles of $Cu^{2+} = M \times V = 0.20 \times 0.50 = 0.10$ mol. Since $Cu^{2+} + 2e^- \rightarrow Cu$, charge $Q = n \times F = 0.10 \times 2 \times 96500 = 19300$ C.

Answer: (B)

Practice Questions (Exam Pattern) — Part 3 (Q11 to Q15)

Q11. How many Faradays of electricity are required for the complete reduction of 1 mole of permanganate ions (MnO_4^-) to manganese(II) ions (Mn^{2+}) in acidic medium?

(A) 1 F **(B)** 3 F
(C) 5 F **(D)** 7 F

Sol. Q11. $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$. The reduction transfers $5e^-$ per formula unit, requiring exactly 5 F (5×96500 C).

Answer: (C)

Q12. During the electrolysis of concentrated aqueous sodium chloride solution (brine) using inert electrodes, the products liberated at cathode and anode are respectively:

(A) Cathode: $Na(s)$; Anode: $O_2(g)$ **(B)** Cathode: $H_2(g)$; Anode: $O_2(g)$
(C) Cathode: $H_2(g)$; Anode: $Cl_2(g)$ **(D)** Cathode: $Na(s)$; Anode: $Cl_2(g)$

Sol. Q12. Water reduces preferentially to yield $H_2(g)$ at cathode. At anode, $Cl_2(g)$ discharges instead of O_2 due to the high **overpotential** of oxygen.

Answer: (C)

Q13. When an aqueous solution of $CuSO_4$ is electrolysed using pure copper electrodes, what reactions occur at the anode and cathode?

(A) Anode: $O_2(g)$; Cathode: $H_2(g)$ **(B)** Anode: $O_2(g)$; Cathode: Cu deposits
(C) Anode: $Cl_2(g)$; Cathode: Cu deposits **(D)** Anode: Cu dissolves; Cathode: Cu deposits

Sol. Q13. With active copper electrodes, copper oxidises at anode ($Cu \rightarrow Cu^{2+} + 2e^-$) and reduces onto cathode ($Cu^{2+} + 2e^- \rightarrow Cu$).

Answer: (D)

Q14. Why does a mercury cell provide a remarkably constant potential (1.35 V) throughout its operational lifetime?

(A) It uses an extremely large excess of zinc-mercury amalgam **(B)** Mercury acts as an inert catalyst that prevents electrode polarisation
(C) It operates as a secondary cell that self-recharges continuously **(D)** The overall cell reaction does not involve any dissolved ions whose concentrations can change

Sol. Q14. Overall reaction: $Zn(Hg) + HgO(s) \rightarrow ZnO(s) + Hg(l)$. No ions appear in the net reaction, so voltage is invariant with time.

Answer: (D)

Q15. Galvanised iron sheets remain immune to corrosion even after the protective zinc layer is scratched because:

(A) Zinc oxide barrier seals the scratch **(B)** Iron is more electropositive than zinc
(C) Zinc forms a passivating alloy with iron **(D)** Zinc has more negative E^0 and acts as sacrificial anode

Sol. Q15. $E^0\left(Z\frac{n^{2+}}{Z}n\right) = -0.76$ V is more negative than $E^0\left(F\frac{e^{2+}}{F}e\right) = -0.44$ V. Zinc oxidises sacrificially, preserving iron cathodic.

Answer: (D)